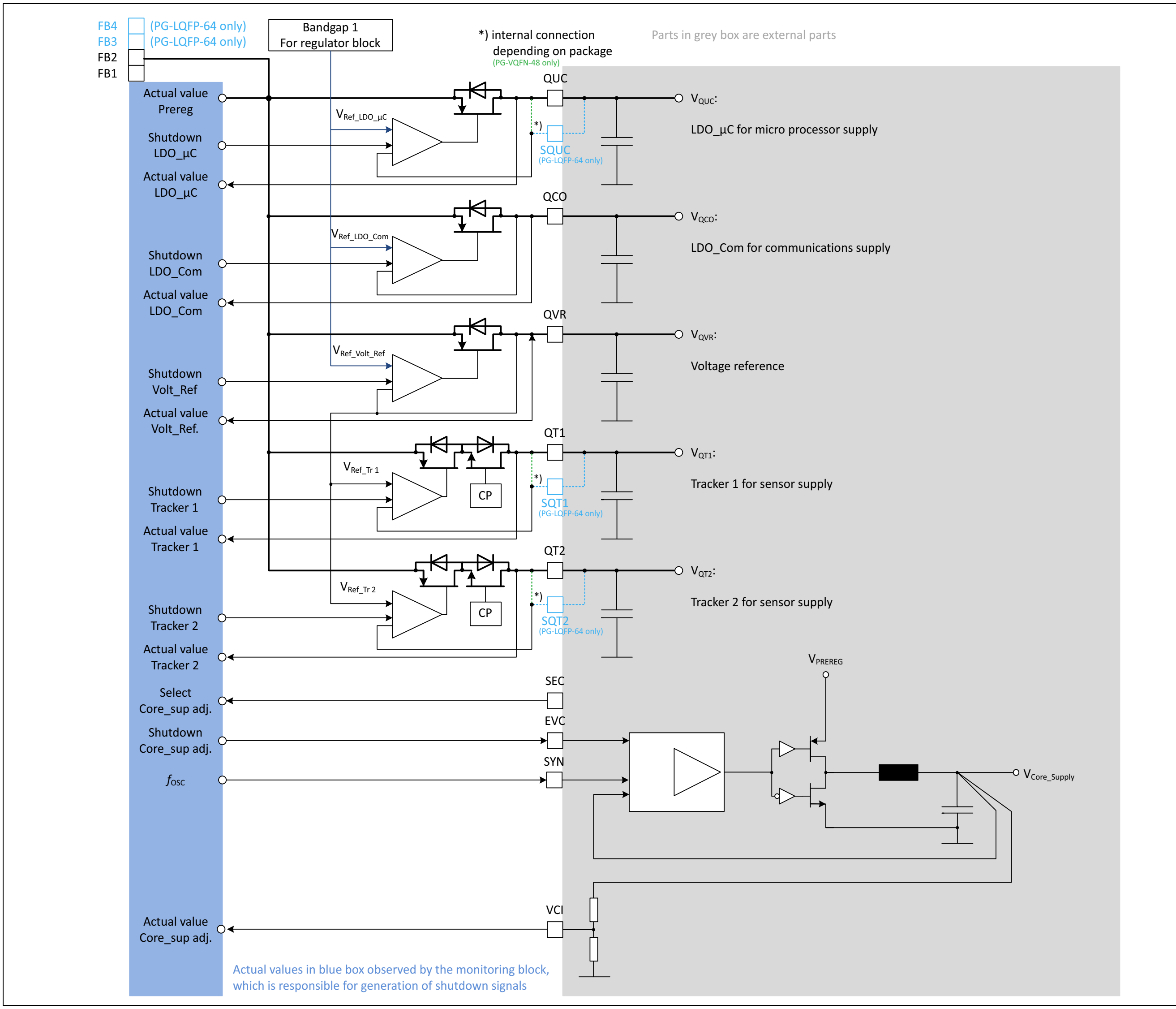


Figure 10 Principle post regulators

7.2 µ-Processor Supply

7.2.1 Functional description

The linear low drop regulator LDO_µC offers a precise 3.3 V or 5.0 V output voltage for micro processor supply.

The regulator is supplied from the intermediate circuit voltage V_{PREREG} which provides a stabilized voltage. The output voltage V_{QUC} (at pin QUC) is controlled by the error amplifier. The actual value is compared to a reference voltage derived from band gap 1 for regulators. The stability of the control loop depends on the load current, the characteristics of the output capacitor and the chip temperature. To ensure a stable operation the output capacitor should be chosen according the specified requirements (capacitance value and electrical series resistance ESR) in **Table 10** "Electrical characteristics". The input capacitor shown in figure below is the output filter capacitor of the step down pre regulator.

Protection circuitry is installed to prevent the regulator and the application from damage:

- To protect the pass element of the LDO_µC from overstress the current limitation will limit the output current to the maximum specified limit. Current sensing is done via a current mirror, no sense resistor is used. In case the maximum current condition is reached, the current will be limited, thus the output voltage will decrease. The regulator is protected against short circuit to ground.
- The output voltage is monitored by the voltage monitoring. In case of over voltage at pin QUC, the LDO_µC will be switched off and the device will move into FAILSAFE state. The event will be stored in the SPI status register (**MONSF1**). In case of under voltage at pin QUC, the device will move into INIT state, pin ROT will be pulled low and the event will be stored in an SPI status register (**MONSF2**). The regulator will not be switched off in case of output under voltage, which is shorter than the short to ground detection time t_{StG}. If the under voltage should be present for more than t_{StG}, the device will move into FAILSAFE state. This event will be stored in an SPI status register as well (**MONSF0**).
- There is a dedicated temperature sensor for this regulator. In case the power stage temperature exceeds the pre warning threshold, an interrupt will indicate this event and it will be stored in an SPI status register (**OTWRNSF**). If the power stage temperature exceeds the temperature shutdown threshold, the device will move into FAILSAFE state, the regulator will be switched off and the event will be stored in an SPI status register (**OTFAIL**). The off time due to temperature shut down will be at least one second.

If the device enters FAILSAFE state the ROT is pulled low and all supplies are switched off.

If the device enters STANDBY state the LDO_µC is switched off

For further details please refer to **Chapter 11 State Machine**.

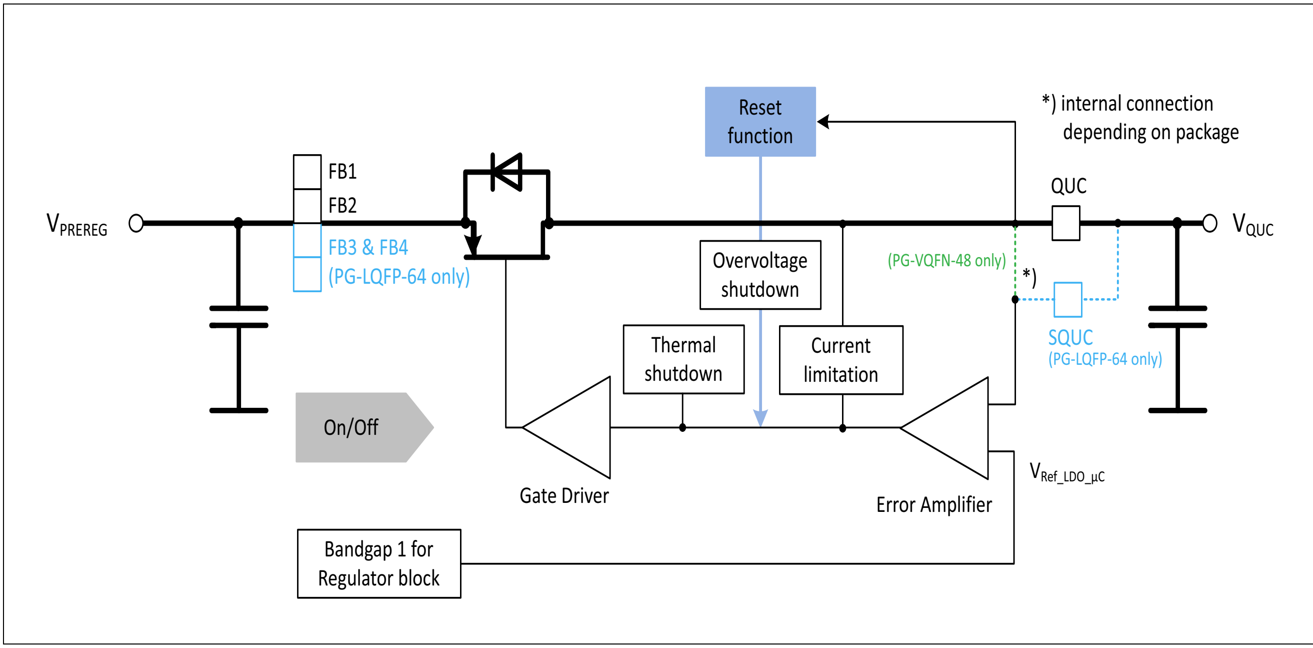


Figure 11 Low drop linear regulator for micro processor supply LDO_µC

7.2.2 Electrical characteristics

Table 10 Electrical characteristics: µProcessor supply

V_{VS} =6.0 V to 40 V, T_j = -40°C to +150°C, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

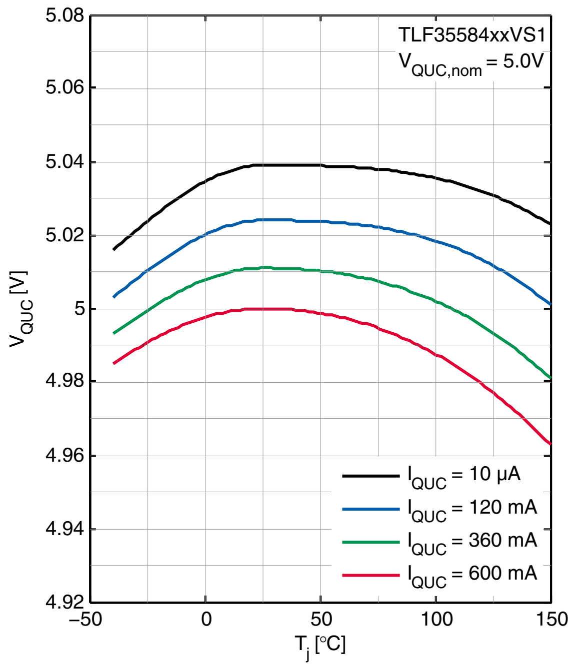
Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
µ Processor supply							
Output voltage TLF35584xxVS1	V_{QUC}	4.9	5.0	5.1	V	0 mA ≤ I_{QUC}≤ 600 mA	P_7.2.2.1
Output voltage TLF35584xxVS2	V_{QUC}	3.23	3.3	3.37	V	0 mA ≤ I_{QUC}≤ 600 mA	P_7.2.2.2
Output current limitation	$I_{QUC, max}$	650	–	1100	mA	–	P_7.2.2.3
Drop voltage TLF35584xxVS1	$V_{dr, QUC}$	–	–	400	mV	[1]	P_7.2.2.4
Drop voltage TLF35584xxVS2	$V_{dr, QUC}$	–	–	500	mV	[1]	P_7.2.2.5
Load regulation TLF35584xxVS1	ΔV_{QUC}	–	45	81	mV	I_{QUC} = 100µA to 600 mA	P_7.2.2.6
Load regulation TLF35584xxVS2	ΔV_{QUC}	–	39	60	mV	I_{QUC} = 100µA to 600 mA	P_7.2.2.7
Power supply ripple rejection	$PSRR_{QUC}$	26	–	–	dB	[2] V_{PREREG} = 5.8 V ; ESRC_{QUC} ≤ 100mΩ	P_7.2.2.8
Output capacitor	C_{QUC}	2.2	–	47	µF	[2]	P_7.2.2.9
Output capacitor, ESR	$ESR\ C_{QUC}$	0	–	200	mΩ	[2]	P_7.2.2.10
Over temperature warning threshold	$T_{j,OT, WRN}$	130	145	160	°C	T_j increasing [2]	P_7.2.2.11
Over temperature shutdown threshold	$T_{j,OT, shutdown}$	175	190	205	°C	T_j increasing [2]	P_7.2.2.12
Over temperature sensor hysteresis	$T_{j,OT, hyst}$	–	10	–	°C	[2]	P_7.2.2.13

1) Dropout voltage is defined as the difference between input and output voltage when the output voltage decreases 100 mV from output voltage measured at V_I = $V_{Q,nom}$ + $V_{dr,max}$ + 100 mV.

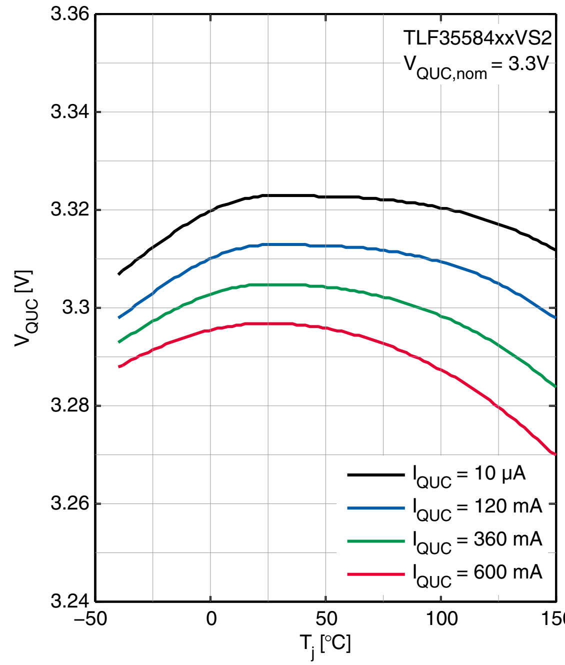
2) Specified by design, not subject to production test

7.2.3 Typical Performance Characteristics

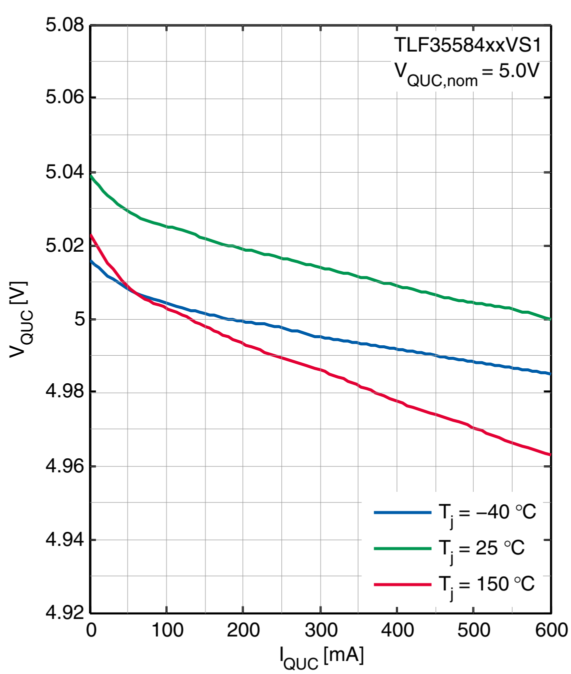
QUC Output Voltage V_{QUC} versus Junction Temperature T_j (TLF35584xxVS1)



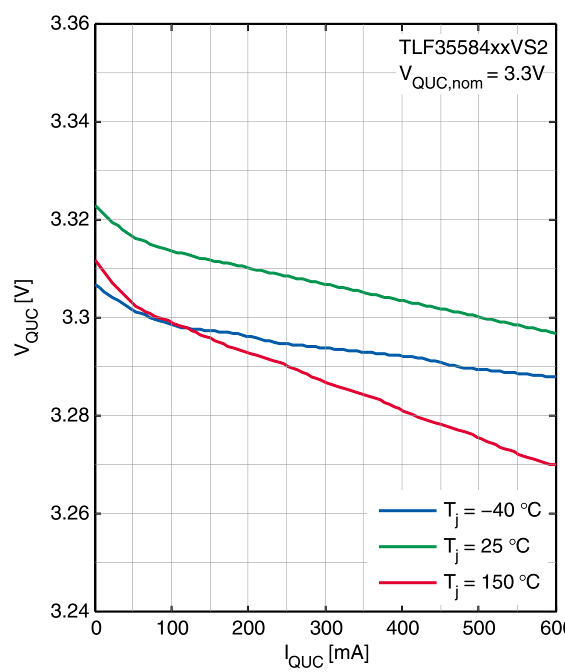
QUC Output Voltage V_{QUC} versus Junction Temperature T_j (TLF35584xxVS2)



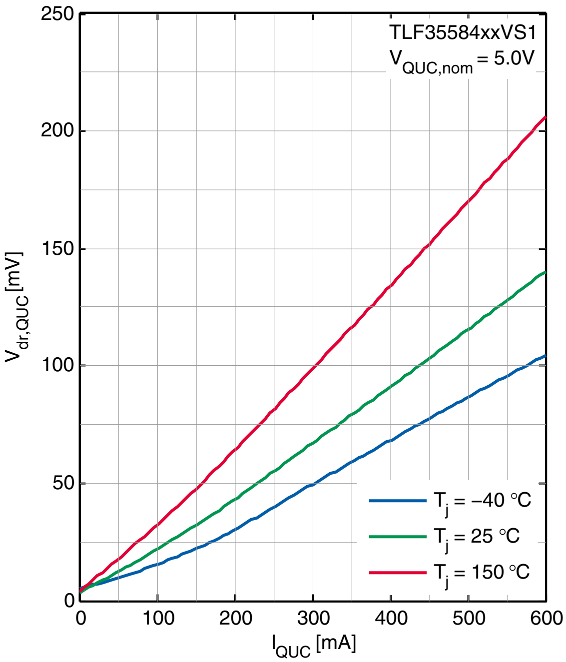
QUC Output Voltage V_{QUC} versus Load Current I_{QUC} (TLF35584xxVS1)



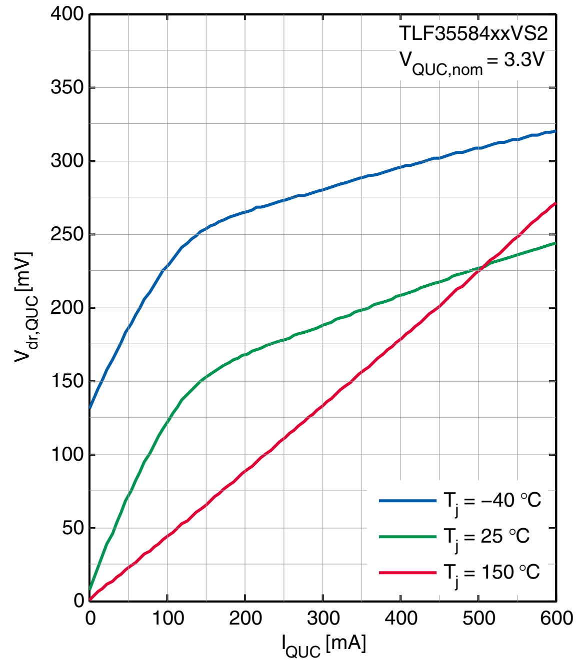
QUC Output Voltage V_{QUC} versus Load Current I_{QUC} (TLF35584xxVS2)



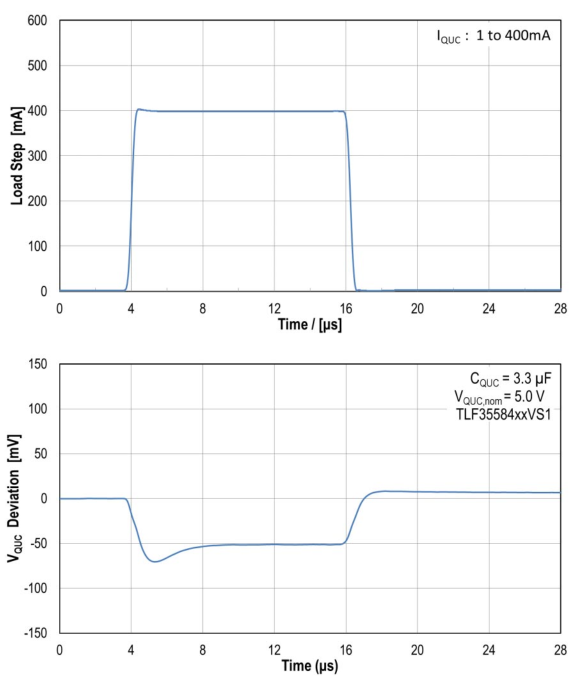
QUC Dropout Voltage $V_{dr,QUC}$ versus Load Current I_{QUC} (TLF35584xxVS1)



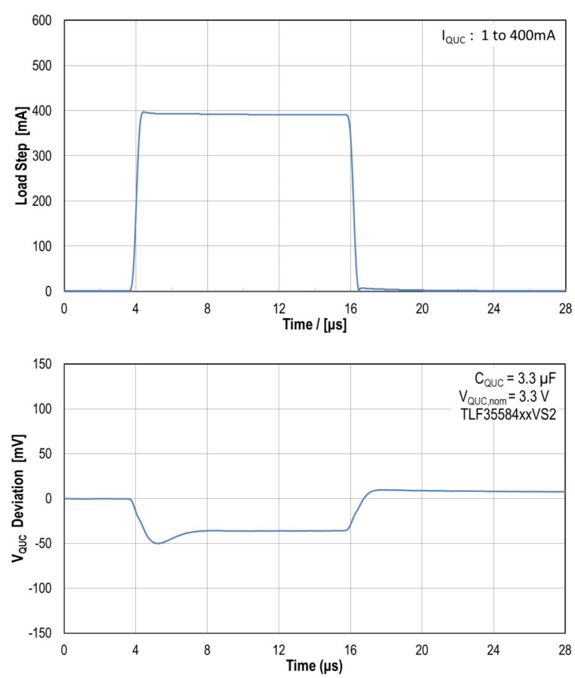
QUC Dropout Voltage $V_{dr,QUC}$ versus Load Current I_{QUC} (TLF35584xxVS2)



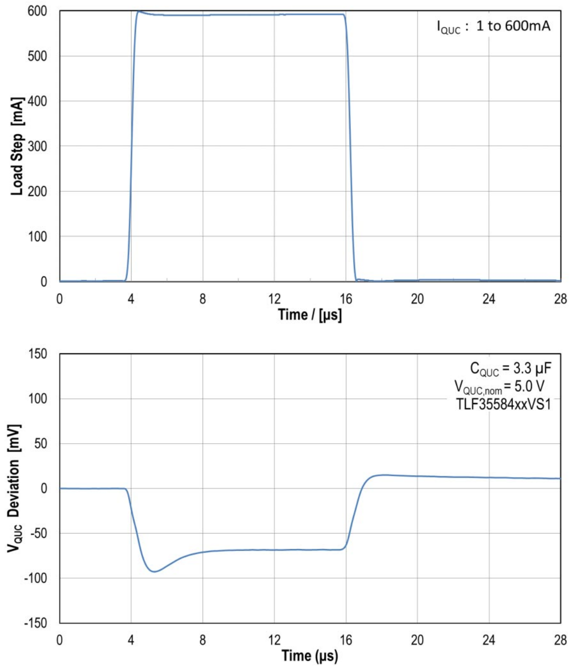
QUC Dynamic Load Response (1mA to 400mA) TLF35584xxVS1 ($V_{QUC,nom}$ = 5.0 V)



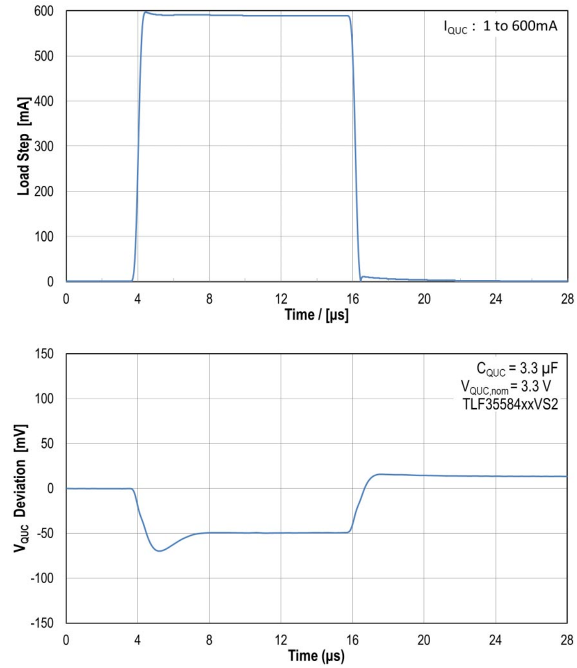
QUC Dynamic Load Response (1mA to 400mA) TLF35584xxVS2 ($V_{QUC,nom}$ = 3.3 V)



QUC Dynamic Load Response (1mA to 600mA) TLF35584xxVS1 ($V_{QUC,nom}$ = 5.0 V)



QUC Dynamic Load Response (1mA to 600mA) TLF35584xxVS2 ($V_{QUC,nom}$ = 3.3 V)



7.3 Communication Supply

7.3.1 Functional description

The linear low drop regulator LDO_Com offers a precise 5.0 V output voltage for communication supply.

The regulator is supplied from the intermediate circuit voltage V_{PREREG}, which provides a stabilized voltage. The output voltage V_{QCO} (at pin QCO) is controlled by the error amplifier. The actual value is compared to a reference voltage derived from band gap 1 for regulators. The stability of the control loop depends on the load current, the characteristics of the output capacitor and the chip temperature. To ensure a stable operation the output capacitor should be chosen according the specified requirements (capacitance value and electrical series resistance ESR) in **Table 11** “Electrical characteristics”. The input capacitor shown in figure below is the output filter capacitor of the step down pre regulator.

Protection circuitry is installed to prevent the regulator and the application from damage:

- To protect the pass element of the LDO_Com from overstress the current limitation will limit the output current to the maximum specified limit. Current sensing is done via a current mirror, no sense resistor is used. In case the maximum current condition is reached, the current will be limited, thus the output voltage will decrease. The regulator is protected against short circuit to ground.
- The output voltage is monitored by the voltage monitoring. In case of over voltage at pin QCO, the LDO_Com will be switched off, the event will be indicated by an interrupt and stored in an SPI status register (**MONSF1**). In case of under voltage at pin QCO, the event will be indicated by an interrupt and stored in an SPI status register (**MONSF2**). The regulator will not be switched off in case of output under voltage, which is shorter than the short to ground detection time t_{StG}. If the under voltage should be present for more than t_{StG}, the regulator will be switched off. This event will be stored in the SPI status register (**MONSF0**) and an interrupt will be generated.
- There is a dedicated temperature sensor for this regulator. In case the power stage temperature exceeds the pre warning threshold, an interrupt will indicate this event and it will be stored in an SPI status register (**OTWRNSF**). If the power stage temperature exceeds the temperature shutdown threshold, this event will be stored in an SPI status register (**OTFAIL**), the regulator will be switched off and an interrupt will be generated. After a temperature shut down the LDO_Com can be re enabled via SPI command.

The regulator LDO_Com is switched off in STANDBY and FAILSAFE state. In INIT, SLEEP, NORMAL and WAKE state LDO_Com is ON or OFF depending on the SPI configuration.

For further details please refer to **Chapter 11 State Machine**.

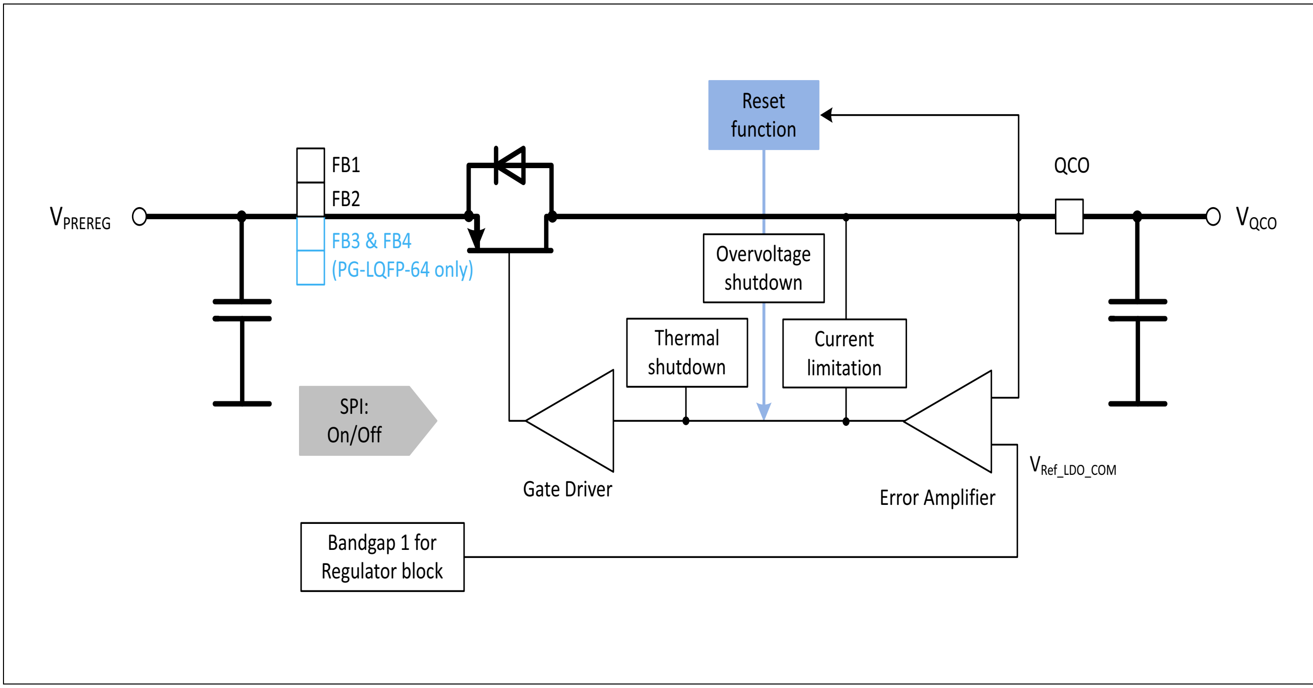


Figure 12 Low drop linear regulator for communications supply LDO_Com

7.3.2 Electrical characteristics

Table 11 Electrical characteristics: Communication supply

V_{VS} =6.0 V to 40 V, T_j = -40°C to +150°C, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

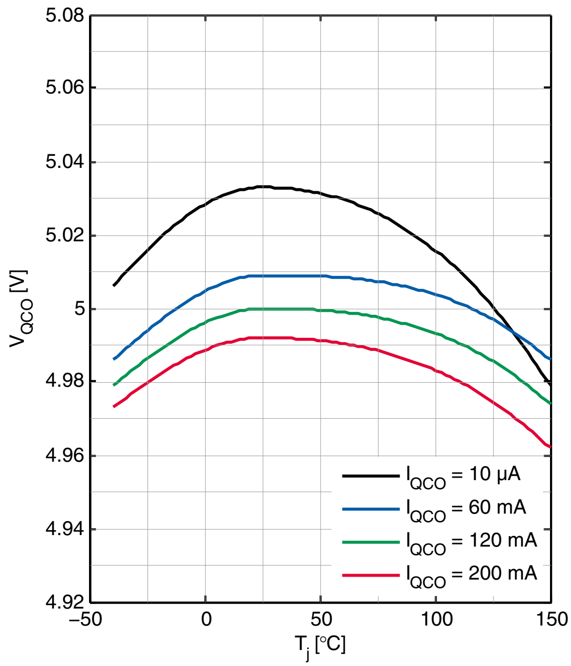
Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Communication supply							
Output voltage	V_{QCO}	4.90	5.00	5.10	V	0 mA ≤ I_{QCO}≤ 200 mA	P_7.3.2.1
Output current limitation	$I_{QCO,\ max}$	250	–	400	mA	–	P_7.3.2.2
Drop voltage	$V_{dr,\ QCO}$	–	–	400	mV	1)	P_7.3.2.3
Load regulation	ΔV_{QCO}	–	40	70	mV	I_{QCO} = 100µ to 200 mA	P_7.3.2.4
Power supply ripple rejection	$PSRR_{QCO}$	26	–	–	dB	2) V_{PREREG} = 5.8 V ; ESRC_{QCO} ≤ 100mΩ	P_7.3.2.5
Output capacitor	C_{QCO}	1	–	47	µF	2)	P_7.3.2.6
Output capacitor, ESR	$ESR\ C_{QCO}$	0	–	200	mΩ	2)	P_7.3.2.7
Over temperature warning threshold	$T_{j,OT,\ WRN}$	130	145	160	°C	T_j increasing 2)	P_7.3.2.8
Over temperature shutdown threshold	$T_{j,OT,\ shutdown}$	175	190	205	°C	T_j increasing 2)	P_7.3.2.9
Over temperature sensor hysteresis	$T_{j,OT,\ hyst}$	–	10	–	°C	2)	P_7.3.2.10

1) Dropout voltage is defined as the difference between input and output voltage when the output voltage decreases 100 mV from output voltage measured at $V_I = V_{Q,nom} + V_{dr,max}$ + 100 mV.

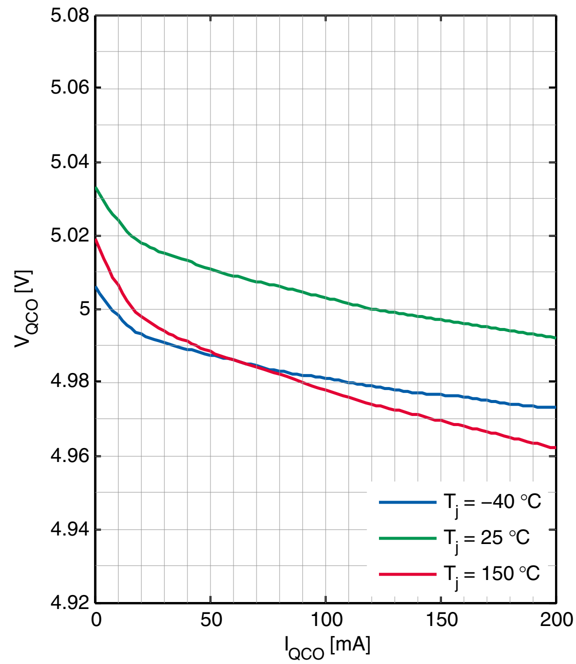
2) Specified by design, not subject to production test

7.3.3 Typical Performance Characteristics

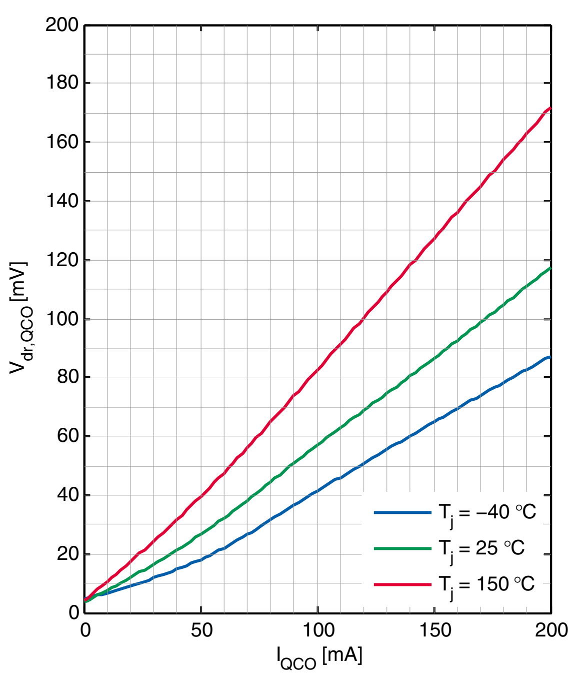
QCO Output Voltage V_{QCO} versus Junction Temperature T_j



QCO Output Voltage V_{QCO} versus Load Current I_{QCO}



QCO Dropout Voltage $V_{dr,QCO}$ versus Load Current I_{QCO}



QCO Dynamic Load Response (1mA to 100mA) ($V_{QCO,nom}$ = 5.0 V)